Question 1 Compaction

An earthfill will be constructed on the field. Spesific weight and void ratio of the borrow area are 2,70 and 0,95 respectively. A laboratory compaction test was conducted on the soil sample (soil sample was recovered from borrow area) and results presented in the table given below

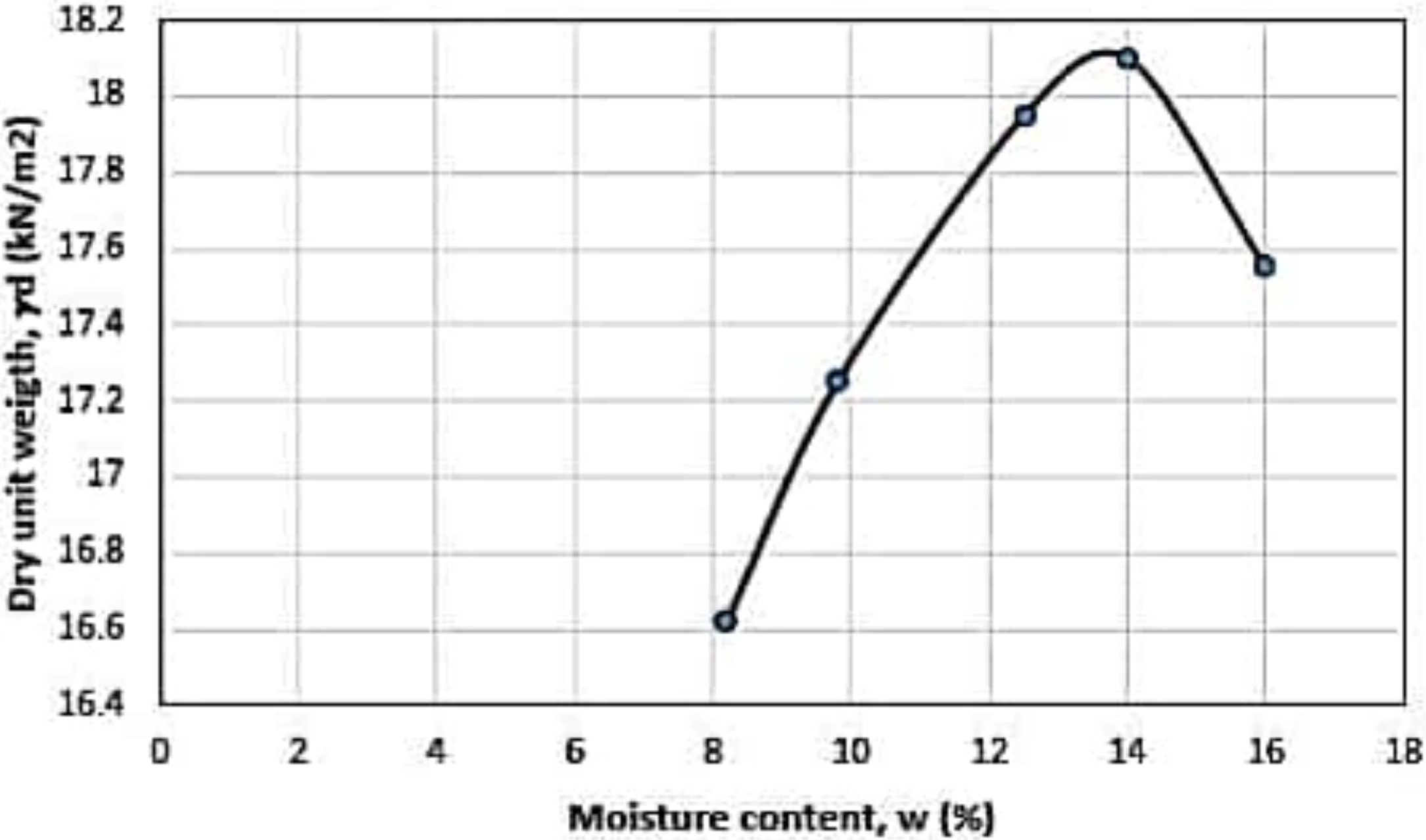
γ_d (kN/m^3)	16,62	17,25	17,95	18,10	17.55
w (%)	8,20	9,80	12,50	14.0	16.0

a) Draw compaction curve and determine w_{opt}.

b) Calculate degree of saturation of the compacted soil at w_{opt}.

c) A soil sample was recovered from compacted fill in the field and water content and natural unit weight were found 12% and 19,6 kN/m^3 respectively. Determine whether the relative compaction meets the requirements or not (RC=95%)

d) If the volume of the excavated soil is to be 30.000m^3 how many m^3 compacted fill can be produced from excavated soil?


18.2
18
17.8
17.6
17.4
17.2
17
16.8
16.6
16.4
Dry unit weigth, γd (kN/m2)
0
2
4
6
8
10
12
14
16
18
Moisture content, w (%)

Solution 1

b) $S.e = G_s.w \Rightarrow e = \frac{G_s.w}{S}$ ①

② $\gamma_d = \frac{\gamma_w.G_s}{1+e} \Rightarrow$ dry unit weight of soil at w_{opt} is 18.1 kN/m³ ($\gamma_{d(max)}$)

$18.1 = \frac{9.81 \cdot 2.7}{1+e} \Rightarrow e = 0.463$

$S = \frac{2.7 \cdot 14}{0.463} \Rightarrow S = \%81.57$

c)

$$\text{Relative Compaction } (R) = \frac{\gamma_{d(field)}}{\gamma_{d(max-lab)}} \times 100$$

$\gamma_{d(field)} = \frac{\gamma_n}{1+w} = \frac{19.6}{1+0.12} = 17.5 \text{ kN/m}^3$

$R = \frac{17.5}{18.1} \times 100 = \%96.7 > \%95$

d) $V = 30000 \text{ m}^3$

$e = 0.95 \quad w = \%12$

$G_s = 2.70 \quad \gamma_n = 19.6 \text{ kN/m}^3$

$G_s = \frac{\gamma_s}{\gamma_w} \Rightarrow 2.70 \rightarrow \gamma_s = 27 \text{ kN/m}^3 \Rightarrow \gamma_s = \frac{M_s}{V_s}$

$e = \frac{V_v}{V_s} \Rightarrow 0.95 = \frac{V_b}{V_s} \Rightarrow V = V_b + V_s = 1.95 V_s$

$1.95 V_s = 30000$

$V_s = 15385 \text{ m}^3$ ★

$V_b = 14615 \text{ m}^3$

$M_s = 27 \times 15385 = 415395 \text{ kN}$, $w = \frac{M_w}{M_s} \rightarrow M_w = 415395 \cdot 0.12 = 49847.4 \text{ kN}$

$\gamma_n = \frac{M_s + M_w}{V_b + V_s} \Rightarrow 19.6 = \frac{49847.4 + 415395}{V_b + 15385} \Rightarrow$

$V_b = 8351.85 \text{ m}^3$

$V_s = 15385$

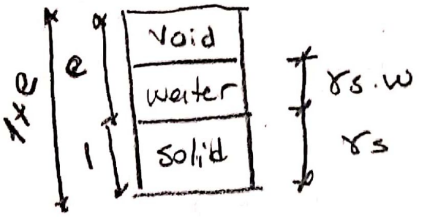
23736 m^3

Question 2

The in situ void ratio of a soil is 0.9 and the specific gravity of soil solids is 2.7. This soil is to be excavated and transported to a construction site for use in a compacted fill. How many cubic meters compacted soil are produced with $w = 12\%$ and $\gamma_n = 19.4\ kN/m^3$ using $25000\ m^3$ of excavated soil?

Solution 2

$$e = \frac{V_v}{V_s},\quad V_T = 1 + e \Rightarrow \left\{ \frac{V_T}{V_s} = \frac{1 + e_{fill}}{1} \right\}$$



$$\gamma_n = \frac{\gamma_s \cdot (1 + w)}{1 + e} \Rightarrow 1 + e_{fill} = \frac{(2.7 \cdot 9.81)\,(1 + 0.12)}{19.4}$$

$$1{,}529 = 1 + e_{fill}$$

→ The void ratio of excavated soil is $e = 0.9$

→ The volume of solid in excavated soil: $V_s = \frac{V_{Total}}{1 + e}$

$$= \frac{25000}{1 + 0.9} \cong 13160\ m^3$$

→ The volume of solid in fill will be same with excavated one.

$$V_{fill} = V_s \cdot (1 + e_{fill}) = 13160 \times (1.529)$$

$$= 20120\ m^3$$

Question 3

A field unit-weight determination test performed on a clayey silt. The moist density and moisture content of compacted soil were determined as 2.1 kg/m^3 and 10%, respectively. Laboratory test results have shown that $\gamma_{d,max} = 19$ kN/m^3. Determine:

a) Dry unit weight of compaction in the field

b) Relative compaction in the field.

Solution 3

a) Moist unit weight of compacted soil $= \frac{(2100) \times (9.81)}{1000}$

$= 20.60$ kN/m^2

$$\gamma_d = \frac{\gamma}{1 + \frac{w(\%)}{100}} = \frac{20.62}{1 + \frac{10}{100}}$$

$= 18.73$ kN/m^3

b) $R = \frac{\gamma_{d(field)}}{\gamma_{d(max.)}} = \frac{18.73}{19} = 98\% > 95\%$ ✓

Question 4

Following are the results of a soil planned to use for an eartfill contruction.

$$w = 10\%, \quad e = 1.2, \quad \text{and} \quad G_s = 2.70$$

For a given soil, following are the results of compaction tests conducted in the laboratory.

$$\gamma_{d(max)} = 18.50 \text{ kN/m}^3$$

$$w_{opt} = 14\% \text{ kN/m}^3$$

After the compaction of the soil in the field, a hole with a volume of 900 cm^3 was hollowed and moist mass and dry mass were determined as 1935 g and 1698 g, respectively. The specification require that: γ_d must be at least 0.95 $\gamma_{d(max)}$, and moisture content (w) should be within $\pm$ % 2 of w_{opt}.

a) Calculate the dry unit weight and moisture content of the fill. Determine whether compaction meets the requirements or not.

b) Calculate the how many m^3 of soil excavated from borrow area if the volume of the fill is to be 20000 m^3

c) If 25 % water loss occurs durney transportation, determine whether water was added to fill or not.

Solution 4

$e_{excavated} = 1.2$

$w_{excavated} = 10\%$

$\gamma_{d(max),Lab} = 18.5\ kN/m^3$

$w_{opt} = 14\%$

a) $\gamma_{d(fill)} = \left(\frac{1698}{900}\right)(9.81) = 18.5\ kN/m^3$ (M_s / V)

$\gamma_{n(fill)} = \left(\frac{1935}{900}\right)(9.81) = 21.09\ kN/m^3$

$R = \frac{\gamma_{d(fill)}}{\gamma_{d(max),Lab}} = \frac{18.5}{18.5} = 100\% > 95\% \Rightarrow$ It satisfies the requirement.

$\gamma_n = \frac{\gamma_n}{1+w} \Rightarrow \frac{21.09}{1+w} = 18.5 \Rightarrow w = 14\%$

$= 12 < 14 < 16$ ✓

b)

$\gamma_{d(excavated)} = \frac{G_s \cdot \gamma_w}{1+e} = \frac{(2.7)\cdot(9.81)}{1+1.2} = 12.04\ kN/m^3$

$\Rightarrow W_{d(excavated)} = W_{d(fill)}$

$\Rightarrow \gamma_{d(excavated)} \times V_{(excavated)} = \gamma_{d(fill)} \times V_{(fill)}$

$\Rightarrow (12.04)\cdot(V_{(ex.)}) = \underbrace{(18.5)(20000)}_{370000} \Rightarrow V_{(ex.)} \simeq 30731\ m^3$ ✓

c)

$W_d = 370000\ kN$

$W_{w(ex.)} = 370000\cdot(0.1) = 37000\ kN$

Water loss in transportation

$37000 - 37000\,(0.25) = 27750\ kN$

$W_{w(fill)} = 370000\cdot(0.14) = 51800\ kN$

Added water in fill soil

$W_{w(added)} = 51800 - 27750 = 24050\ kN \xrightarrow[9.81]{} 2452\ m^3$ ✓